EMC Lab 2: TEM line

Laboratory Report

Fatima Hasan
Student ID: 2888172h

October 22, 2026

Contents

1 Introduction **2**

2 Results and Analysis **2**

7.1 Travelling Wave : Wavelength Measurement 2

8.2 Lissajous Figures: Phase and Wavelength Measurement 3

9.3 Standing Waves: Magnetic Field (Inductive Probe) 5

10.4 Standing Waves: Electric Field Measurement 7

11 Discussion **9**

12 Conclusion **10**

1 Introduction

In this experiment, the electric and magnetic field distributions of high frequency electromagnetic waves travelling on a 50 Ω coaxial transmission line at a frequency of 750 MHz are investigated. The spatial distributions of the electric field E_r and the magnetic field H_θ are directly measured using a slotted line with capacitive and inductive probes under matched load, open circuit, and short circuit termination conditions.

The coaxial transmission line supports the Transverse Electromagnetic (TEM) mode, where both electric and magnetic field vectors are perpendicular to the direction of propagation. The matched load termination condition produces a travelling wave, whereas standing waves are formed under the open circuit and short circuit termination conditions. In these two cases, nodes and antinodes are established along the line. The electric field E_r and the magnetic field H_θ are measured using a sampling oscilloscope with a maximum operating frequency of 1 GHz. Sinusoidal waveforms and Lissajous patterns are used to measure the wavelength.

2 Results and Analysis

7.1 Travelling Wave : Wavelength Measurement

Measurement	Start Position (m)	End Position (m)	λ Measured (m)
1	49 cm	10 cm	= 39 cm

Table 1: Measured positions and calculated wavelength for the travelling wave

Theoretical Wavelength:

The expected wavelength is calculated using:

$$\lambda = \frac{c}{f} \tag{1}$$

Substituting values:

$$\lambda = \frac{c}{f} = \frac{3 * 10^8}{750 * 10^6} = 0.40m = 40cm \tag{2}$$

where:

- λ wavelength (m)
- $c = 3 \times 10^8$ m/s speed of light in free space.
- f frequency (Hz)

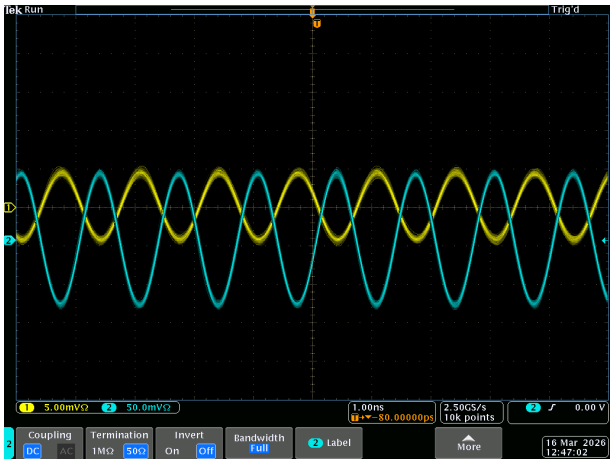


Figure 1: The carriage at 49 cm observed on the oscilloscope.

Percentage Error:

$$\% \text{ Error} = \left| \frac{\lambda_{\text{measured}} - \lambda_{\text{expected}}}{\lambda_{\text{expected}}} \right| \times 100\% \tag{3}$$

Substituting values:

$$\% \text{ Error} = \left| \frac{40 - 39}{40} \right| \times 100\% = 2.5\ \% \tag{4}$$

Comment:

The measured wavelength is **39 cm**; which is closely related to the theoretical free space wavelength of **40 cm** at **750 MHz**. This gives a percentage error of only **2.5%**. This small discrepancy may be attributed to a slight mechanical reading error.

8.2 Lissajous Figures: Phase and Wavelength Measurement

Position z (cm)	**Phase Angle ϕ (°)**	**Lissajous Shape**
0	52	Ellipse, negative slope (wide)
4	21.8	Ellipse, negative slope (narrow)
8	8.8	Straight line (minimum)
12	34.5	Ellipse, positive slope (narrow)
16	54.4	Ellipse, positive slope (wide, near circular)
20	50.8	Ellipse, negative slope (wide)
24	21.4	Ellipse, negative slope (narrow)
28	7.8	Straight line (minimum)
32	32.1	Ellipse, positive slope (narrow)
36	51.8	Ellipse, positive slope (wide)
40	46.9	Ellipse, positive slope (wide)

Table 2: Lissajous figure observations and phase angle measurements along the transmission line.

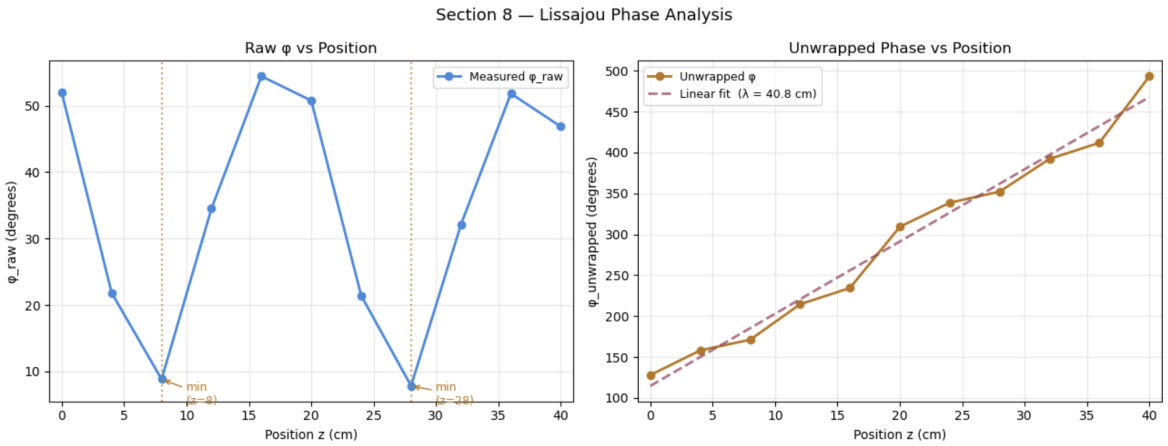


Figure 2: Lissajous figure plot.

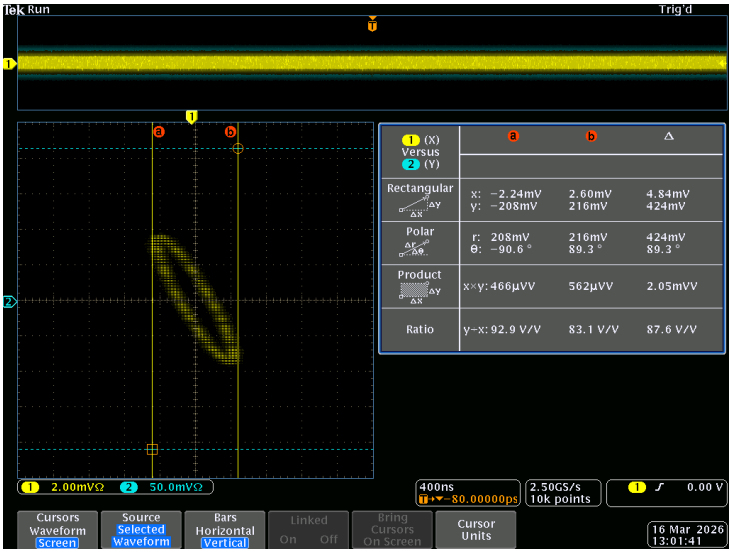


Figure 3: Lissajous figure observed on the oscilloscope.

Note: The highlighted rows indicate where the Lissajous ellipse collapsed to a straight line through the origin which corresponds to a phase difference of π radians between the reference and travelling wave signals.

Wavelength from Minima Separation

The distance between two consecutive minima corresponds to half a wavelength. Therefore:

$$\lambda = 2 \times \Delta z_{\text{minima}} = 2 \times (28 - 8) = 2 \times 20 = \boxed{40 \text{ cm}} \tag{5}$$

Method	Section	λ Measured	Error (%)
Sinusoidal trace (peak alignment)	7	39.0 cm	2.5
Lissajous minima spacing	8	40.0 cm	0.0

Table 3: Wavelength extraction using three independent methods.

Comment:

The Lissajou approach provided a result of 40 cm which exactly matched the theory; whereas the sinusoidal approach provided a result of 39 cm as calculated in Section 7 (with a 2.5% error). This is because the Lissajou approach inherently provides a more accurate result, as determining when the shape collapses into a straight line at a minimum provides a more precise visual cue than trying to determine when two sinusoids have a phase shift of 2π by eye which can be prone to human error.

9.3 Standing Waves: Magnetic Field (Inductive Probe)

#	OC Position (cm)	OC Type	SC Position (cm)	SC Type	Δz (cm)
1	2.4	min	2.1	max	0.3
2	12.6	max	12.4	min	0.2
3	22.1	min	23.4	max	1.3
4	32.6	max	32.2	min	0.4
5	42.1	min	42.9	max	0.8
Mean	—	—	—	—	**0.6**

Table 4: Measured maxima and minima positions for Open Circuit (OC) and Short Circuit (SC) terminations , magnetic field probe.

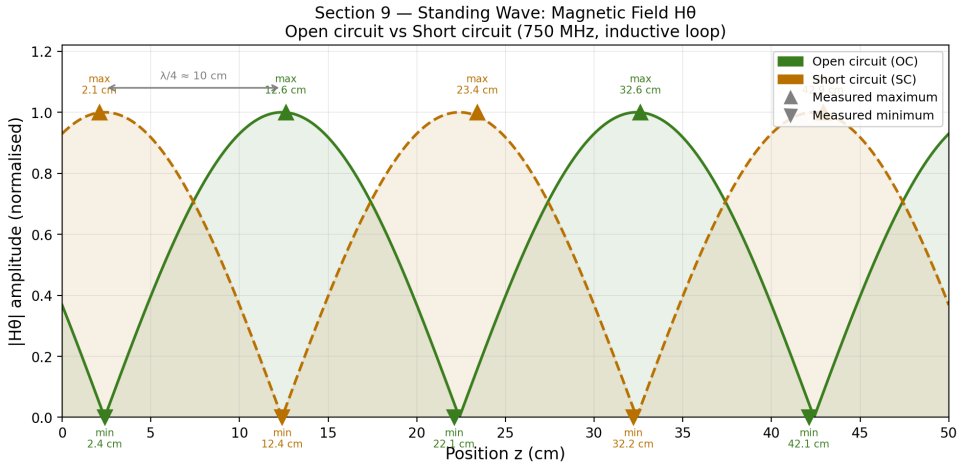


Figure 4: Magnetic field standing wave pattern along the transmission line.

Wavelength Extraction:

The wavelength is calculated from the separation between two consecutive minima (or maxima) in the standing wave pattern:

$$\lambda = 2 \times \Delta z_{\text{minima}} = 2 \times (22.1 - 2.4) = 2 \times 19.7 = \boxed{39.4 \text{ cm}} \tag{6}$$

Confirmation of $\lambda/4$ Shift:

Switching from *open circuit* to *short circuit* termination is expected to shift the entire standing wave pattern by a quarter wavelength. The expected shift is:

$$\frac{\lambda}{4} = \frac{40.0}{4} = \boxed{10.0 \text{ cm}} \tag{7}$$

The average positional difference between OC and SC features is only $\overline{\Delta z} = 0.6$ cm confirming that each maximum became a minimum and vice versa at approximately the same location consistent with a $\lambda/4$ pattern displacement.

Comment:

The transition from open circuit termination to short circuit termination resulted in every maximum becoming a minimum and vice versa at approximately the same position on the line, with an average positional difference of only $\overline{\Delta z} = 0.6$ cm. This is further confirmation that the standing wave pattern has been shifted by $\lambda/4 \approx 10$ cm. as theory predicts that the pattern shifts by one quarter wavelength.

10.4 Standing Waves: Electric Field Measurement

#	OC Position (cm)	OC Type	SC Position (cm)	SC Type	Δz (cm)
1	0.0	max	10.8	max	10.8
2	10.8	min	20.6	min	9.8
3	20.8	max	30.9	max	10.1
4	30.1	min	40.6	min	10.5
5	39.8	max	48.8	max	9.0
Mean	—	—	—	—	**10.04**

Table 5: Measured maxima and minima positions for Open Circuit (OC) and Short Circuit (SC) terminations electric field probe.

Wavelength Extraction:

The wavelength is calculated from consecutive maxima spacing in the *open circuit* case:

$$\lambda_1 = 2 \times (20.8 - 0.0) = \boxed{41.6 \text{ cm}} \tag{8}$$

$$\lambda_2 = 2 \times (39.8 - 20.8) = \boxed{38.0 \text{ cm}} \tag{9}$$

Taking the average:

$$\lambda_{\text{avg}} = \frac{41.6 + 38.0}{2} = \boxed{39.8 \text{ cm}} \tag{10}$$

The percentage error with respect to the theoretical value is:

$$\% \text{ Error} = \left| \frac{39.8 - 40.0}{40.0} \right| \times 100 = \boxed{0.5\%} \tag{11}$$

Confirmation of $\lambda/4$ Shift:

The expected quarter wavelength shift is:

$$\frac{\lambda}{4} = \frac{40.0}{4} = 10.0 \text{ cm} \tag{12}$$

The measured average shift was $\overline{\Delta z} = 10.04$ cm giving a % error of:

$$\% \text{ Error} = \left| \frac{10.04 - 10.0}{10.0} \right| \times 100 = \boxed{0.4\%} \tag{13}$$

Unlike the magnetic field case in Section 9. the transformation from open circuit termination to short circuit termination preserved the feature type at each position, as maxima remained maxima and minima remained minima while the pattern shifted by $\lambda/4$.

Spatial Quadrature: E_r vs H_θ (OC Termination):

Feature	E_r Position (cm)	H_θ Position (cm)	Type	Separation (cm)
1	0.0	2.4	E max $\leftrightarrow$ H min	2.4
2	20.8	22.1	E max $\leftrightarrow$ H min	1.3
3	39.8	42.1	E max $\leftrightarrow$ H min	2.3
4	10.8	12.6	E min $\leftrightarrow$ H max	1.8
5	30.1	32.6	E min $\leftrightarrow$ H max	2.5
Mean	—	—	—	**2.06**

Table 6: Comparison of E_r and H_θ feature positions under open circuit termination.

The theoretical value for this is 0 cm, so that the maxima of E_r and H_θ coincide.

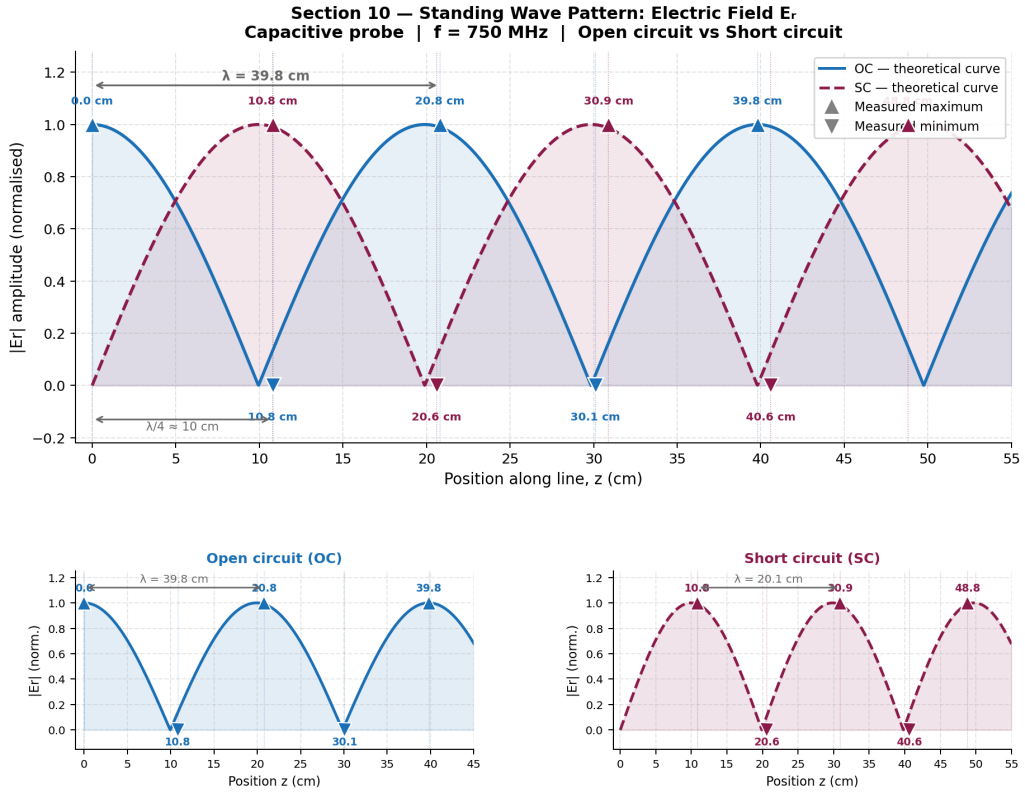


Figure 5: Electric field standing wave pattern for openc ircuit and short circuit terminations, with H_θ overlaid for quadrature comparison.

Comment:

Using the capacitive probe a standing wave pattern was observed for E_r under both terminations. For the open circuit case maxima were observed at positions 0, 20.8, and 39.8 cm. This gave an average wavelength of $\lambda = 39.8$ cm (0.5% error), unlike the magnetic field case in Section 9, changing from open circuit to short circuit maintained the same feature type for each position. In other words maxima remained max. However, the entire pattern was shifted by an average of 10.04 cm (0.4% error compared to $\lambda/4 = 10.0$ cm).

Comparing the positions of the electric and magnetic fields in the open circuit termination the max of E_r were found to coincide closely with the minima of H_θ, with an average spacing of only 2.06 cm ,which confirms the theoretical spatial quadrature relationship that:

$$E_r \propto \cos(\beta z) \qquad H_\theta \propto \sin(\beta z) \tag{14}$$

where the two fields are spatially displaced by $\lambda/4$.

11 Discussion

The results for all four sections show good agreement with the theoretical predictions. The wavelength given a frequency of $f = 750$ MHz, was consistently measured between 39 and 40 cm across all sections compared with the theoretical free space value of:

The maximum percentage error was therefore 2.5%, confirming that the TEM mode propagates on the coaxial line at the speed of light.

In Section 7, the sinusoidal trace method gave $\lambda = 39$ cm (2.5% error), while the Lissajous method in Section 8 gave $\lambda = 40$ cm (0% error). The increased accuracy of the latter method is consistent with the principle that null detection is more accurate than peak detection by eye due to the sharper definition of the ellipse as it collapses to a straight line.

The standing wave measurements in Sections 9 and 10 were also confirming the theoretical behaviour for OC and SC terminations. The OC to SC shift was consistently around 10 cm for both sections in good agreement with the theoretical $\lambda/4$ shift of 10 cm. An important difference was observed between the two fields For the magnetic field switching termination caused the maxima and minima to swap places at locations close to $\lambda/2$. For the electric field; the pattern simply shifted by $\lambda/4$ with all feature types preserved, exactly as expected from the theoretical field expressions for the open circuit termination:

$E_r \propto \cos(\beta z) \qquad H_\theta \propto \sin(\beta z)$

The experiment has successfully demonstrated the TEM mode of propagation in the coaxial transmission line and the theoretical field distributions under matched, open, and short circuit conditions.

Sources of Experimental Error

The experimental inaccuracies are attributed to the following sources:

1. **Precision of the mechanical carriage readings:** the scale of the carriage has limited resolution introducing systematic reading errors of approximately 1–2 mm per measurement.
2. **Probe coupling:** the capacitive and inductive probes slightly disturb the field being measured perturbing the locations of the maxima and minima.
3. **Signal reflections:** the connector interfaces are not ideal, producing parasitic reflections that alter the standing wave pattern from its theoretical form.

4. **Oscilloscope reading:** accurate identification of maxima and minima on the oscilloscope screen is subject to visual judgement error particularly in the case of the Lissajous figures.

Overall the experiment successfully demonstrated the TEM mode of propagation on a coaxial transmission line and validated the theoretical field distributions under matched, open, and short circuit conditions.

12 Conclusion

The TEM mode of propagation has been successfully demonstrated for the coaxial transmission line in this experiment. In addition, the theoretical field distributions for matched, Open Circuit, and Short Circuit conditions have been successfully demonstrated. The wavelength was measured using four different methods and was found consistent across all four. The maximum error was 2.5% compared to the theoretical value of 40.0 cm. The $\lambda/4$ shift between the open circuit and short circuit standing wave patterns was verified for the electric and magnetic fields. In addition, the spatial quadrature relationship $E_r \perp H_\theta$ was experimentally verified with an average positional deviation of 2.06 cm.